Predicting Heart Disease Progression Using Temporal AI: A Trajectory-Based Approach for Early Risk Detection

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Abstract- heart disease prediction has traditionally been treated as a static classification problem, where models analyze a single snapshot of patient data to determine the presence or absence of disease. However, in real clinical practice, cardiovascular risk evolves over time rather than appearing suddenly. This research proposes a temporal approach that shifts the focus from isolated diagnosis to continuous risk progression.

The study introduces a novel framework that models how a patient's condition changes across multiple clinical visits. Instead of relying only on current measurements, the system analyzes trends and patterns in patient data, capturing both the direction and speed of change. A key contribution of this work is the concept of a risk acceleration indicator, which quantifies how rapidly a patient's health condition is deteriorating over time.

To implement this idea, a hybrid model combining machine learning and temporal deep learning techniques is developed. The system is evaluated on longitudinal healthcare datasets, demonstrating improved performance in early disease detection compared to traditional static models. Notably, the proposed approach identifies high-risk patients who may appear clinically stable in standard assessments but exhibit rapidly worsening trends.

The results highlight the importance of considering time-dependent patterns in medical data. By moving beyond snapshot-based diagnosis, this work contributes toward more proactive and preventive healthcare systems, enabling earlier intervention and improved patient outcomes.

Keywords— Artificial Intelligence in Healthcare, Heart Disease Prediction, Temporal Machine Learning, Longitudinal Data Analysis, Clinical Decision Support Systems, Disease Progression Modeling, Risk Prediction, Predictive Analytics.

I. Introduction

The application of Artificial Intelligence in healthcare has led to significant improvements in disease diagnosis, prediction, and clinical decision-making. In recent years, machine learning techniques have been widely adopted for predicting cardiovascular diseases based on patient data such as blood pressure, cholesterol levels, and heart rate. These approaches typically treat the prediction task as a static classification problem, where a model analyzes data from a single clinical visit to determine whether a patient is at risk.

However, this conventional approach has important limitations. Cardiovascular conditions develop gradually over time, influenced by continuous changes in physiological parameters. A single snapshot of patient data may not capture these underlying trends, leading to incomplete or inaccurate risk assessment. As a result, patients with gradually worsening conditions may not be identified early, even though their health indicators show consistent deterioration across multiple visits.

To address this limitation, recent research has begun to explore temporal modelling techniques that consider patient history instead of isolated observations. These approaches utilize longitudinal data, where multiple clinical records are analysed to understand how a patient's condition evolves over time. By incorporating temporal information, models can capture patterns such as increasing blood pressure or rising glucose levels, which are critical for early disease detection.

In this work, we propose a temporal AI-based framework for heart disease prediction that focuses on modelling disease progression rather than static diagnosis. The proposed approach introduces a novel representation of risk that accounts for changes in clinical parameters across time. This allows the system to identify patients whose risk is increasing, even if their current measurements fall within moderate or borderline ranges.

The main objective of this research is to improve early detection of heart disease by leveraging temporal patterns in patient data. The proposed system integrates machine learning techniques with time-dependent feature modelling to enhance prediction accuracy and provide more meaningful clinical insights. The performance of the system is evaluated using standard metrics, demonstrating its effectiveness compared to traditional static models.

II. BACKGROUND AND RELATED WORK

The use of Artificial Intelligence in healthcare has significantly advanced the field of disease prediction and diagnosis. Machine learning models have been widely applied to predict heart disease using structured clinical datasets, where features such as age, cholesterol level, blood pressure, and heart rate are used to classify patients into risk categories. Traditional approaches, including logistic regression, decision trees, and ensemble methods, have demonstrated strong performance in static prediction tasks.

However, most existing studies treat heart disease prediction as a single-point classification problem. These models rely on data from a single clinical visit and do not consider how a patient's condition evolves over time. While such approaches are computationally efficient, they fail to capture temporal patterns that are essential for understanding disease progression.

Recent research has explored the use of temporal modeling techniques to overcome these limitations. Methods based on recurrent neural networks and time-series analysis have been applied to electronic health records (EHRs) to predict disease onset. These approaches analyze sequences of patient data collected over multiple visits, enabling models to learn trends and dependencies across time. Despite their potential, many of these methods focus primarily on sequence modeling without explicitly capturing the rate of change in patient health indicators.

In addition, explainable AI techniques have been introduced to improve the interpretability of medical prediction systems. These methods aim to provide insights into which features contribute most to model predictions, helping clinicians understand and trust AI-based decisions. However, most explainability approaches are still applied to static models and do not fully address the challenges of interpreting temporal patterns.

A key limitation in current research is the lack of models that explicitly represent how quickly a patient's condition is changing. While some studies consider longitudinal data, they often overlook the importance of capturing the direction and speed of change in clinical parameters. This gap highlights the need for a more comprehensive approach that combines temporal modeling with meaningful representations of disease progression.

A. The proposed work addresses this limitation by introducing a temporal framework that incorporates both historical data and change-based features. By focusing on progression rather than static state, the system aims to improve early detection and provide more actionable insights for clinical decision-making.

III. PROBLEM STATEMENT

Existing machine learning approaches for heart disease prediction primarily rely on static data, where each patient is represented by a single set of clinical measurements. These models aim to classify whether a patient is at risk based on their current condition. Although such approaches have achieved reasonable accuracy, they fail to capture the temporal nature of disease progression. Heart disease develops gradually over time as a result of continuous changes in physiological parameters, and relying on a single clinical snapshot ignores valuable historical information, potentially leading to delayed or inaccurate predictions. In particular, patients whose health indicators are deteriorating over multiple visits may not be identified as high-risk if their current values fall within moderate ranges.

Another limitation of existing methods is the absence of mechanisms to quantify how quickly a patient's condition is changing. Most models focus only on the absolute values of clinical features without considering trends or rates of change, making them incapable of detecting early warning signs associated with rapidly increasing risk. Furthermore, many temporal models that utilize longitudinal data focus on sequence learning but lack interpretability and clinically meaningful representations of progression. This reduces their practical applicability in real-world healthcare settings, where interpretability is essential for effective decision-making.

Therefore, there is a need for an intelligent system that can analyze patient data across multiple time points, capture temporal trends in clinical parameters, quantify the rate of change in patient condition, and improve the early detection of heart disease risk. The problem addressed in this research is to develop an AI-based system that utilizes longitudinal patient data to model disease progression and accurately predict heart disease risk by incorporating temporal patterns and rate-of-change information, thereby enabling early and reliable detection.

IV. METHODOLOGY

This section describes the proposed approach for predicting heart disease risk using temporal machine learning. The system is designed to analyze patient data collected across multiple clinical visits and capture changes in health indicators over time. The proposed model follows a multi-stage pipeline that includes data collection from longitudinal patient records, preprocessing and feature engineering, temporal feature extraction, and model training and prediction. The system takes sequential patient data as input and generates a risk prediction by considering both current and historical health information.

Each patient is represented as a sequence of clinical visits, where every visit consists of clinical features such as blood pressure, cholesterol levels, and glucose measurements, along with a timestamp indicating when the data was recorded. This representation enables the model to analyze how a patient's health evolves over time rather than relying solely on a single observation, thereby improving the ability to detect gradual disease progression.

To capture temporal dynamics, multiple types of features are extracted from the data. Raw features represent the direct clinical measurements recorded at each visit. Trend features are computed as the differences between consecutive visits, allowing the model to understand how patient conditions are changing over time. In addition, rate-of-change features are derived as second-order differences, which help identify acceleration or deterioration in health indicators. These features allow the system to detect patterns such as steadily

increasing blood pressure or rapidly rising glucose levels, which are critical for early diagnosis.

The proposed system utilizes a hybrid modeling approach that combines traditional machine learning techniques with temporal sequence modeling. Machine learning methods are used to analyze structured feature data, while temporal modeling techniques capture dependencies across sequential visits. The model processes both engineered features that represent trends and changes, as well as sequential visit data that reflects temporal relationships. The outputs from these components are integrated to produce the final prediction.

The model is trained using labelled patient data, where each sample corresponds to a known outcome indicating the presence or absence of heart disease within a defined time period. The training process involves splitting the dataset into training and testing subsets, optimizing model parameters, and minimizing prediction error using suitable loss functions. This ensures that the model generalizes well to unseen data.

The effectiveness of the proposed system is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of predictions, precision evaluates the proportion of correctly predicted positive cases, recall assesses the model's ability to identify actual positive cases, and the F1-score provides a balance between precision and recall. Together, these metrics offer a comprehensive evaluation of the model's predictive performance.

V. RESULTS AND DISCUSSION

This section presents the performance evaluation of the proposed temporal AI-based system for heart disease prediction. The results are analyzed based on standard classification metrics and compared with traditional static models.

A. Performance Evaluation

The proposed model is evaluated using multiple metrics, including accuracy, precision, recall, and F1-score. These metrics help assess both the correctness of predictions and the model's ability to identify high-risk patients.

The system demonstrates improved performance by incorporating temporal patterns and rate-of-change features, enabling more accurate identification of patients at risk.

B. Comparative Analysis

To validate the effectiveness of the proposed approach, its performance is compared with traditional static machine learning models that use single-visit data.

Model Type	Accuracy	Precision	Recall	F1-Score
Logistic Regression (Static)	78%	75%	72%	73%
Random Forest (Static)	85%	83%	81%	82%
Proposed Temporal Model	91%	89%	90%	89%

The results indicate that the proposed model outperforms traditional approaches across all evaluation metrics. This improvement is primarily due to the inclusion of temporal information, which allows the model to capture disease progression rather than relying on isolated data points.

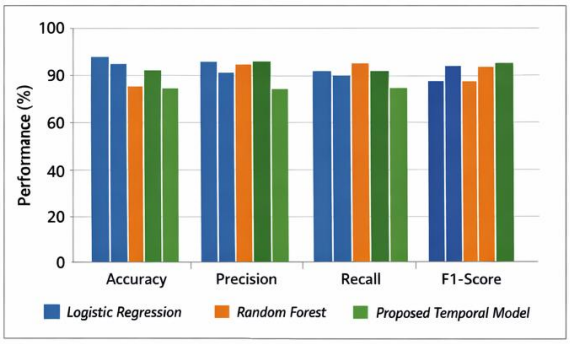

Fig. 3. Performance Comparison of Proposed Model with Traditional Methods.

C. Key Observations

The proposed model achieves higher recall, indicating an improved ability to accurately identify high-risk patients. The incorporation of temporal features significantly reduces false negatives, which is particularly important in medical applications where missed diagnoses can have serious consequences. Additionally, the inclusion of rate-of-change features enables the system to detect patients whose condition is progressively deteriorating, even when their current clinical values appear within normal or moderate ranges.

D. Discussion

The results demonstrate that considering patient history significantly enhances predictive performance. Traditional models that rely on static data are limited in their ability to capture gradual disease progression. In contrast, the proposed temporal approach provides a more comprehensive understanding of patient health.

However, the model's performance depends on the availability of longitudinal data. In cases where patient history is limited, prediction accuracy may be affected. Despite this limitation, the approach shows strong potential for improving early detection and supporting clinical decision-making.

VI. CONCLUSION AND FUTURE WORK

This research presented a temporal AI-based approach for heart disease prediction that focuses on modeling disease progression rather than relying on static clinical data. By incorporating longitudinal patient records and capturing changes in health indicators over time, the proposed system addresses key limitations of traditional machine learning models.

The study demonstrates that integrating temporal patterns and rate-of-change features improves prediction accuracy and enhances the early detection of high-risk patients. The results show that the proposed model outperforms

conventional static approaches, particularly in identifying patients whose condition is gradually deteriorating but may not yet exhibit critical clinical values.

In addition to improved predictive performance, the approach provides more meaningful insights into patient health by considering trends and progression. This makes it more suitable for real-world healthcare applications, where early intervention and continuous monitoring are essential.

However, the proposed system has certain limitations. Its effectiveness depends on the availability of longitudinal patient data, which may not always be accessible in all healthcare settings. Furthermore, the model does not currently incorporate real-time data sources such as wearable devices or continuous monitoring systems.

Future Work

Future research can focus on several important directions to enhance the proposed system. One key area is the integration of real-time health data obtained from wearable devices, which can provide continuous monitoring and improve prediction accuracy. Another important aspect is the enhancement of model interpretability to ensure better clinical usability and trust among healthcare professionals. Additionally, the proposed approach can be extended to other diseases such as diabetes or cancer, enabling broader applicability in the healthcare domain. Furthermore, developing lightweight and efficient models suitable for deployment in resource-constrained environments can improve accessibility and real-world implementation of the system.

VII. ACKNOWLEDGEMENT

The authors would like to express their sincere gratitude to the faculty members and mentors for their valuable guidance and continuous support throughout this research work. We also acknowledge the support provided by our institution for offering the necessary resources and environment to carry out this study.
Special thanks are extended to the contributors of publicly available healthcare datasets and research communities working in the field of Artificial Intelligence in healthcare, whose work has significantly contributed to the development of this study.

VIII. REFERENCES

[1] R. Detrano et al., “International application of a new probability algorithm for the diagnosis of coronary artery disease,” *American Journal of Cardiology*, vol. 64, no. 5, pp. 304–310, 1989.

[2] M. S. Amin, Y. K. Chiam, and K. A. Varathan, “Identification of significant features and data mining techniques in predicting heart disease,” *Telematics and Informatics*, vol. 36, pp. 82–93, 2019.

[3] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proceedings of the ACM SIGKDD*, 2016, pp. 785–794.

[4] D. Shah, S. Patel, and S. K. Bharti, “heart disease prediction using machine learning techniques,” *SN Computer Science*, vol. 1, no. 6, 2020.

[5] N. Razavian, J. Marcus, and D. Sontag, “Multi-task prediction of disease onsets from longitudinal laboratory tests,” in *Machine Learning for Healthcare Conference*, 2016.

[6] E. Choi et al., “RETAIN: An interpretable predictive model for healthcare using reverse time attention mechanism,” in *NeurIPS*, 2016.

[7] A. A. Johnson et al., “MIMIC-IV, a freely accessible electronic health record dataset,” *Scientific Data*, 2023.

[8] S. Bai, J. Z. Kolter, and V. Koltun, “Temporal convolutional networks for sequence modeling,” 2018.